Tutorial 4

Elementary matrices, singularity and invertibility

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1. Consider the matrix

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 5 & 4 & 5 \\ 2 & 0 & 3 \end{pmatrix}.$$

(a) Use the algorithm in the notes for Gaussian elimination to get A into row echelon form.

(b) Is A invertible? Use your answer to (a) only to give a reason for your answer. (Do not try to find the inverse!)

(c) Now find the **reduced** row-echelon form of A.

[Hint: Use the row-echelon form in (a) and then carry on in a suitable way to make all the pivots equal to 1 and get 0's above the diagonal as well, if possible.]

(d) For every elementary row operation e you have used, write down the corresponding elementary matrix $E = e(I_3)$.

(e) Use your answer to (d) to find elementary matrices E_1, E_2, E_3 and E_4 such that $E_4\ E_3\ E_2\ E_1 A = I_3$. Make sure you get the E's in the correct order.

(f) Write A as a product of elementary matrices. [Hint: Look at (e).]

(g) Write A^{-1} as a product of elementary matrices.

[*Hint:* How do we get the inverse of a product of invertible matrices?]

2. Let $A = \begin{pmatrix} 0 & 4 & 1 \\ 2 & 7 & 0 \\ -4 & -6 & 2 \end{pmatrix}$.

(a) Find elementary matrices E_1, E_2 and E_3 and a matrix U in row-echelon form such that $E_3 E_2 E_1 A = U$.

(b) Is the matrix U you obtained non-singular? Give a reason for your answer.

(c) Can A be written as a product of elementary matrices? Give a reason for your answer.

3. Let $A = \begin{pmatrix} 1 & 0 & -2 \\ -3 & 1 & 2 \\ 2 & -3 & 4 \end{pmatrix}$.

   (a) Use the algorithm introduced in class to determine whether A is invertible, and if it is, find A^{-1}.

   Now answer the following two questions without doing any further Gaussian elimination, using only your answer to (a).

   (b) Find the solution set for the system of equations $A\mathbf{x} = 0$.

   (c) Find the solution set for the system of equations $A\mathbf{x} = \mathbf{b}$ where $\mathbf{b} = \begin{pmatrix} 4 \\ 0 \\ -2 \end{pmatrix}$.

4. Let A be an $m \times n$ matrix, and suppose it is row reduced to a matrix U in row echelon form. Note that A is **not** necessarily a square matrix, and that U is a row-echelon form of A, not of an augmented matrix. Are the following statements true or false? If true, give a proof. If false, give a counter-example.

   [**Warning:** The Invertibility Theorem is a theorem about **square** matrices.]

   (a) If there is a pivot in every column of U, then the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for every $\mathbf{b}$.

   (b) If the equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for every $\mathbf{b}$, then the solution (for each $\mathbf{b}$) is unique.

   (c) If the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for *some* $\mathbf{b}$, then it has a unique solution for *every* $\mathbf{b}$.

   (d) The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for *every* $\mathbf{b}$ if and only if U has no zero rows.

   (e) Now answer each question again under the assumption that A **is square**.

5. Use the **definitions** and the **theorems** about invertible matrices in section 1.5 and 1.6 to prove the following:

   (a) If B is invertible, then B^{-1} is invertible.

   [*Hint:* Your proof need not be longer than one line. Find the inverse of B^{-1}; look at the definition of invertibility to see what you need for B^{-1} to be invertible.]

   (b) If AB is non-singular, then B is non-singular.

   [*Hint:* Use the definition of a non-singular matrix.You may find it easier to prove the contra-positive of this statement.]

(c) If A and B are square matrices and AB is invertible, then both A and B are invertible. [*Hint:* Use your answers to (a) and (b).]

(d) If A and B are square matrices such that $AB = I$, then $BA = I$. (From this it follows that in the definition of the inverse of a square matrix A, it is enough to require that there be a square matrix B such that $AB = I$.)

[Hint: Use (c) and the fact that I is invertible.]

(e) Now write down the converse of the statement in 5(b) and prove or disprove it.

6. *This question gives you an idea of how we'll test your understanding of proofs.*

Read the following proof and then answer the questions following it.

Proof. Let A be an $n \times n$ non-singular matrix. We show that A is row equivalent to I_n. We know that A is row equivalent to a matrix U in row echelon form. (1)
The matrix U must look like this:

$$\begin{pmatrix} u_{11} & u_{12} & \dots & u_{1n} \\ 0 & u_{22} & \dots & u_{2n} \\ 0 & 0 & \dots & \\ \vdots & & \dots & \vdots \\ 0 & 0 & \dots & u_{nn} \end{pmatrix}.$$

If we can show that the u_{ii} are non-zero for $1 \le i \le n$ then we can carry on row reducing until we obtain I_n. (2)
Suppose that one of the u_{ii}'s is zero. Then, because U is a square matrix in row echelon form, u_{nn} must be zero too. (3)
But that means that we have at most $n-1$ pivots, so at least one free variable in the equation $U\mathbf{x} = \mathbf{0}$. (4)
It follows from this that the equation has infinitely many solutions. (5)
But since U and A are row equivalent, we have by a previous result that $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions. (6)
This is a contradiction of our assumption, so we must have $u_{ii} \neq 0$ for every i. (7)
Since we now know that U can be row-reduced to I_n, the result is proved. □

(a) Carefully state the result proved here.

(b) Why can we make the statement at (1)?

(c) Explain why it is possible to perform elementary row operations on U which will make all the entries above u_{ii} zero provided $u_{ii} \neq 0$. How does it follow from this that we can row-reduce U to I_n?

(d) Why can we say, at (3), that $u_{nn} = 0$?

(e) Why can we say, at (4), that there are at most $n-1$ pivots?

(f) Explain why the fact that there is a free variable in the equation $U\mathbf{x} = \mathbf{0}$ means that the equation will have infinitely many solutions.

(g) State the "previous result" used at (6).

(h) What is the contradiction referred to at (7)?

(i) Explain why the result is proved once we know U can be row-reduced to I_n.